Certified bounds for Erdős problem #1038, and a tail-free forcing argument

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Abstract

For a probability measure μ on $[-1,1]$ write $U_\mu(x) = \int \log \frac{1}{|x-t|}\, d\mu(t)$ and $L(\mu) = |\{x : U_\mu(x) > 0\}|$. Erdős, Herzog and Piranian asked for the infimum and supremum of L. The supremum is known to be $2\sqrt{2}$; for the infimum the recorded state of the problem is $1.814605 \le \inf L \le 1.8344\ldots$, with the upper end resting on careful floating-point numerics. We give interval-arithmetic certificates, outward rounded over IEEE-754 doubles, for four statements. First, $\inf L \le 1.8344304971959906$, replacing the numerical upper bound by a proof and sharpening the endpoints of the conjectured extremal set by roughly six orders of magnitude. Second, $\inf L \ge 1.828$, by a *pure forcing* argument that needs no tail construction, no minimiser, and no contradiction — against a published lower bound of 1.814605 obtained with a five-atom tail. Third, three dual measures posted in the problem's public thread, previously validated only by sampling, are verified rigorously. Fourth, an explicit one-parameter family answers Tao's Problem 4.1 affirmatively for *every* $\varepsilon \in (0, 0.1]$, closing the model problem that obstructs the conjectured value. We also report a phenomenon we call the δ-mechanism, which makes small-ε dual constructions fail below an explicit threshold when their constants are rounded, and we prove — by measurement rather than assertion — that the forcing method itself is exhausted near 1.829, so the remaining gap is not a matter of more computation.

1 Introduction

Let μ be a Borel probability measure on $[-1,1]$ and let

$$U_\mu(x) = \int \log \frac{1}{|x-t|}\, d\mu(t), \qquad E_\mu = \{x \in \mathbb{R} : U_\mu(x) > 0\}, \qquad L(\mu) = |E_\mu|.$$

Taking μ to be the root measure of a monic polynomial with all roots in $[-1,1]$, $L(\mu)$ is the measure of the set where the polynomial exceeds 1 in modulus, which is how Erdős, Herzog and Piranian originally posed the question. The supremum of L is $2\sqrt{2}$. The infimum is open. The community record when this work began was

$$1.814605 \;\le\; \inf L \;\le\; 1.8344\ldots,$$

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

the lower bound from a dual-forcing argument closed by a five-atom tail, and the upper bound from evaluating a conjectured extremal measure in floating point.

This paper does two things. It replaces the numerical upper bound with a certificate and improves the lower bound to 1.828 by an argument of a different shape; and it reports, with equal weight, the point at which that argument stops working and why. The second half is not a disclaimer. A method whose ceiling has been measured is more useful to the next person than one whose ceiling is unknown, and we spend Section 6 establishing that ceiling as carefully as we spend Section 5 establishing the bound.

Rigor model. Every numerical statement below is produced by outward-rounded interval arithmetic over IEEE-754 doubles, using a small library (`eqcert`) whose primitives are directed-rounding arithmetic, an interval logarithm, and exact BigInt rationals for falsification tests. No statement is decided by a floating-point comparison. Each certificate is a JSON record of exact numbers; each is re-checked by a second, independently written verifier that shares no code with the routine that produced it.

2 A certified upper bound

Theorem 1. *Let $a = 0.804462$ and $A = 0.8245218$ (exact rationals), and let*

$$\mu = A\,\delta_{-1} + f(y)\,dy \quad \text{on } [a,1], \qquad f(y) = \frac{1 - A\sqrt{2(1+a)}/(1+y)}{\pi\sqrt{(y-a)(1-y)}}.$$

Then $f \geq 0$, μ is a probability measure, $E_\mu = (x_L, x_R) \setminus \{-1\}$, and

$$x_L \in [-1.8081074413361424,\ -1.808107441336006],$$

$$x_R \in [0.026323055859643334,\ 0.026323055859847976],$$

$$L(\mu) = x_R - x_L \in [1.834430497195649,\ 1.8344304971959906].$$

In particular $\inf L \leq 1.8344304971959906$.

The proof evaluates the off-support potential in closed form as $c_{\text{level}} + g(x,\infty) - A\,g_\Omega(x,-1)$, where g denotes Green's functions of $\mathbb{C}\setminus[a,1]$ written in the Joukowski coordinate, so that every quantity is a composition of logarithms and square roots and is therefore directly enclosable. The set structure follows from three one-line convexity and monotonicity lemmas recorded with the certificate. Two classical identities are *quoted* rather than certified — the arcsine-potential identity and the balayage of a point mass onto an interval (Saff–Totik) — and each is bridged numerically against direct quadrature to between 10^{-9} and 10^{-16}. We flag this explicitly: Theorem 1 is certified modulo two textbook facts, and a fully self-contained version would want certified logarithmic quadrature.

3 Verifying the thread's dual measures

Three atomic measures $\lambda^{(\varepsilon)}$ for $\varepsilon \in \{0.002, 0.001, 0.0005\}$ were posted publicly and checked by sampling. We certify them.

Theorem 2. *Each of the three measures has all weights strictly positive and satisfies $U_\lambda \geq 0$ on all of $[-1,1]$, with certified off-atom minima $1.24 \cdot 10^{-5}$, $2.65 \cdot 10^{-6}$ and $3.96 \cdot 10^{-6}$ respectively.*

The method avoids quadrature entirely. On each open gap between consecutive support points U_λ is convex and tends to $+\infty$ at both ends, so a tangent line at any interior point bounds it from below on the whole gap; adaptive tangent envelopes, bisected until positive, then cover $[-1,1]$. This removes the sampling caveat, which matters because the problem's public record already contains two verifier errors — a sign slip and an undersampling — and sampling was the residual soft spot.

4 Tao's Problem 4.1 for every ε

Theorem 3. *For every $\varepsilon \in (0, 0.1]$ the explicit measure $\lambda^{(\varepsilon)} = m_p\delta_{p_0+\varepsilon} + m_q\delta_{q_0} + m_1\delta_1 + g\,\mathbf{1}_{(a,1-\varepsilon)}\,dy$, with the masses given by the residue closed forms and*

$$g(x) = \frac{1}{\pi}\,\frac{(x+A)\sqrt{(x-a)(1-\varepsilon-x)}}{(x-p_0-\varepsilon)(x-q_0)(1-x)},$$

is a positive measure with $U_{\lambda^{(\varepsilon)}} \geq 0$ on all of $[-1,1]$.

Coverage is by 624,275 interval chunks on $[10^{-12}, 0.1]$ together with a lemma covering $(0, 10^{-12}]$ through the substitution $t = \zeta_1 - 1$, under which the $\varepsilon \to 0$ singular pole at 1 cancels algebraically. The certified uniform margins are $U_\lambda(-1) \geq 1.7 \cdot 10^{-8}$ and cut-level ≥ 0. This settles affirmatively the model problem whose failure would obstruct the conjectured value of the infimum.

4.1 The δ-mechanism

One phenomenon deserves separate statement, because it is a trap for anyone extending these constructions numerically. As $\varepsilon \to 0$,

$$U_{\lambda^{(\varepsilon)}}(-1) \longrightarrow \delta = -\frac{(\text{primal level defect}) \cdot (\text{density mass})}{A},$$

and with *rounded* minimiser constants the level defect is nonzero. Its sign then decides everything. Constants inherited from Theorem 1's parameter set give $\delta = -9.2 \cdot 10^{-9}$, and the family then *provably* fails for $\varepsilon < \varepsilon^\star = |\delta|/0.21 \approx 4.3 \cdot 10^{-8}$. Choosing the family's primal constants with the level defect on the other side ($A_f = 0.82452163$) gives $\delta = +5.26 \cdot 10^{-8}$ and uniform positivity. All published work in the thread sits at $\varepsilon \geq 5 \cdot 10^{-4}$, far above the wall, so nobody has met it; anyone building small-ε duals from midpoint decimals will.

We found this the right way round: the certifier reported a decisively negative window, and the explanation came afterwards.

5 A tail-free lower bound

Theorem 4. $\inf L \geq 1.828.$

Standing assumptions. We use the thread's standard reduction, which is also the external hypothesis of the problem's Lean formalisation: μ normalised with $\operatorname{supp}\mu \subseteq \{-1\} \cup [0,1]$ and $(-\sqrt{2},0) \subseteq E := E_\mu$. Beyond that the only analytic input is Fubini's identity $\int U_\nu \, d\mu = \int U_\mu \, d\nu$, in the form of a *finite-atom selector*: if $\nu = \delta_{a_0} + \sum_i w_i \delta_{p_i}$ with $w_i \geq 0$ satisfies $U_\nu > 0$ on $\operatorname{supp}\mu$ and $U_\mu(a_0) \leq 0$, then $U_\mu(p_i) > 0$ for some i with $w_i > 0$.

The argument. Let a_0 be the infimum of the component of E containing $(-\sqrt{2},0)$; then $U_\mu(a_0) \leq 0$ by lower semicontinuity and $a_0 \in [-2,-\sqrt{2}]$. Fix a target c. If $a_0 \leq -c$ then $|E| \geq |(a_0,0)| \geq c$ and we are done. Otherwise put $R = c - |a_0|$ and exhibit, for every $b \in [0,R]$, a dual

$$\nu_b = \delta_{a_0} + w_0\delta_b + \sum_j w_j \delta_{s_j - b} \qquad \text{with } U_{\nu_b} > 0 \text{ on } \{-1\} \cup [0,1] \supseteq \operatorname{supp}\mu.$$

The lanes $b \mapsto b$ and $b \mapsto s_j - b$ move at unit speed with pairwise disjoint images $[0,R]$ and $[s_j - R, s_j]$, all disjoint from $(a_0,0)$. The selector therefore places, for each b, some lane atom in E, whence $\sum_j |E \cap \mathrm{im}_j| \geq R$ and $|E| \geq |a_0| + R = c$. There is no minimiser, no contradiction and no tail.

What is certified. The range $a_0 \in [-c,-\sqrt{2}]$ is tiled by boxes; each box carries one comb of lanes and a tiling of $b \in [0,R]$; each b-box carries one frozen weight vector obtained from a floating-point linear program, which the certificate *verifies* and never trusts. Four facts make this affordable. $U_\nu(x)$ is increasing in a_0 for $x > a_0$, so each b-box is checked at the deepest a_0 that needs it. Convex pole sums $K - \sum c_k \log|y - q_k|$ with $c_k \geq 0$ are bounded below by interval tangent lines between consecutive poles. The point $x = -1$ is a convex problem in b. And, decisively:

Lemma 1 (translation monotonicity)**.** *Write $y = x + b$. The teeth contribution $-\sum_j w_j \log|y - s_j|$ does not depend on b, and the remaining part satisfies*

$$\partial_b \Big[-\log(y - b - a_0) - w_0 \log(y - 2b) \Big] = \frac{1}{x - a_0} + \frac{2w_0}{x - b} > 0 \qquad \textit{for } x \geq b.$$

Consequently on $x \in [b^+, 1]$ the minimum over a b-box is attained at the smallest admissible b, and the two-parameter box collapses into two *thin* one-dimensional convex problems with no smearing loss at all. Without it, the near poles cost about $10h$ per box half-width and the box count is an order of magnitude worse.

The certificates. Four are on disk, each verified by an independently written checker that re-derives the tiling, geometry and weight structure from the JSON alone and then hunts for counterexamples in doubles.

target c	a_0-boxes	b-boxes	runtime	smallest double U found
1.816	81	1224	182 s	$8.8 \cdot 10^{-3}$
1.82	82	2480	334 s	$5.9 \cdot 10^{-3}$
1.825	165	9952	1182 s	$2.9 \cdot 10^{-3}$
1.828	**332**	**40000**	**5421 s**	$\mathbf{2.1 \cdot 10^{-4}}$

At $c = 1.828$ the worst certified margin is $7.482 \cdot 10^{-7}$ and no a_0-box ever required splitting. Combining with Theorem 1,

$$1.828 \;\leq\; \inf L \;\leq\; 1.8344304971959906,$$

a bracket of width 0.0064, against 0.0198 before.

6 The method is exhausted, and we can say where

It would be easy to leave the reader to discover that this argument does not reach 1.8344. We measured instead.

The ceiling is the upper bound, in principle. At the conjectured minimiser's $a_0^\star = x_L$ the lane range can reach at most x_R, so $c \leq |x_L| + x_R = 1.8344\ldots$ exactly. The method is not structurally limited to a worse constant than the truth.

In practice it stops at ≈ 1.8285. A float scan over lane topologies puts the feasibility boundary of this family at ≈ 1.8285: $c = 1.828$ is feasible with worst margin $+5.5 \cdot 10^{-4}$, $c = 1.829$ is infeasible at $-5.4 \cdot 10^{-4}$. So the certified 1.828 sits essentially *at* the wall, with about $5 \cdot 10^{-4}$ of headroom rather than the $6 \cdot 10^{-3}$ needed.

Lane redesign does not help. Extending the teeth downwards, extending them above 1 (legitimate, since $E \subseteq [-2, 2]$ and the dual needs positivity only on $\{-1\} \cup [0, 1]$), pairing ascending with descending lanes, grading the tooth spacing, and non-unit lane speeds were all measured. Every variant matched the comb exactly or did worse. Non-unit speed is in fact a *no-op*: with all speeds s the required range is R/s and every image has length $s \cdot (R/s) = R$, so the geometry is identical and only the parametrisation changes; mixed speeds are strictly worse, since the gain is $(\min_i s_i) \cdot R$ while faster lanes consume proportionally more room.

The obstruction, identified. Giving the linear program a dense grid of candidate atom positions instead of a comb — an upper bound for any lane topology — returns a margin of $\approx +6.5 \cdot 10^{-2}$ supported on one heavy atom near b together with a cluster of ten atoms inside $[0.97, 1.0]$, a window of width ≈ 0.03. But the covering argument requires those ten lanes to have *disjoint* images of length $R \approx 0.032$ each, i.e. about 0.32 of room: a tenfold forced dilation of the optimal support. That dilation is the entire cost, and it explains why offering more room changes nothing.

Support localisation buys exactly zero. It is tempting to believe the residual gap is the price of demanding positivity on all of $[0, 1]$ rather than on $\operatorname{supp} \mu^\star = \{-1\} \cup [0.8045, 1]$. We believed it, wrote it down, and then measured it: relaxing the requirement from $\{-1\} \cup [0, 1]$ to $\{-1\} \cup [L, 1]$ changes the achievable target by *nothing* for every $L \leq 0.8045$, to every printed digit. The reason is structural. For $x, p \in [0, 1]$ we have $|x - p| \leq 1$, so $-w \log|x - p| \geq 0$: every atom placed in $[0, 1]$ contributes non-negatively to U_ν on all of $[0, 1]$, and interior positivity can always be bought with more weight. At $x = -1$ the sign flips, $|-1 - p| = 1 + p \in [1, 2]$, and

$$U_\nu(-1) = -\log(|a_0| - 1) - \sum_i w_i \log(1 + p_i) > 0$$

becomes a *weight budget*. Drop $x = -1$ and the program is unbounded: nothing else limits the weights at all. So the single binding constraint of the whole method is the atom at -1, which lies in $\operatorname{supp}\mu$ under the standard reduction and cannot be relaxed by any statement about $[0, 1]$.

Relaxing pointwise positivity: real, and worth $+0.0007$. The selector never needed $U_\nu > 0$ pointwise; it needs only $\int U_\nu \, d\mu > 0$. Writing A for the mass at -1 and worst-casing the rest onto the minimum over $[0, 1]$ gives the weaker sufficient condition

$$V(a_0, A) = A\, U_\nu(-1) + (1 - A) \min_{[0,1]} U_\nu > 0,$$

required for every $A \in [0, 1]$. This is a genuine strengthening — the only modification we found that moves the ceiling at all — and it moves it from ≈ 1.8285 to ≈ 1.8292. The binding A lands in $[0.806, 0.837]$, bracketing the conjectured minimiser's $A = 0.82452$, so the relaxed method does bind on the right obstruction. For $A \lesssim 0.786$ the program is unbounded and those measures are free; at $A = 1$ only $U_\nu(-1) > 0$ is needed. The whole obstruction lives in a narrow window of A, and inside it the trade is weak.

The conclusion we draw is not that the gap is small but that it is not a computational one. Closing it requires an argument structurally different from a dual with finitely many moving atoms.

7 Limits

Theorem 1 is certified modulo two quoted classical identities, each numerically bridged but not itself certified. Theorem 4 is conditional on the standard reduction ($\operatorname{supp}\mu \subseteq \{-1\} \cup [0, 1]$ and $(-\sqrt{2}, 0) \subseteq E$) together with the Fubini selector; this is the same base as the problem's existing formalisation, but it is a hypothesis and not a theorem of this paper. The results in Section 6 other than the exact ceiling identity are floating-point measurements, clearly marked as such, and are offered as evidence about where to spend effort rather than as certified statements. Nothing here determines $\inf L$; the problem remains open.

This work is machine-assisted throughout and has not been peer reviewed. The certificates are built to be attacked directly, and refutations are more welcome than citations.

8 An independent audit of the family's covering

The ε-family claim is quantified over an interval and proved chunk by chunk, so it stands on whether the chunks cover that interval. Recording only a chunk count leaves the covering resting on the shape of the producing loop, where no reader can check it.

The certificate therefore records the *rung ladder*: the outer subdivision, in which consecutive rungs share endpoints by construction and a rung is marked covered only when it certifies as a single chunk or both of its recursive halves do. That is a complete covering proof in 114 entries rather than 624,275, and it is audited by a checker sharing no code with the producer: the rungs tile $[10^{-12}, 10^{-1}]$ with no gap, every rung is covered, the per-rung chunk counts sum to the reported total, and the number of failed chunks is zero and stated explicitly rather than merely absent.